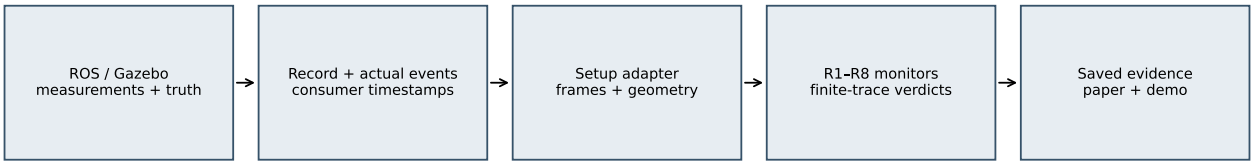
Recorded Verification of UAV–UGV Navigation

Abstract

This report evaluates recorded behavior through reusable, acquisition-aware requirement monitors. Simulator adapters supply independent geometry and actual consumer/actuator timing. Missing evidence remains inconclusive; unmeasured requirements remain unavailable. The results are finite sampled-trace evidence, not continuous safety or discrete supervisor proofs.

1 System and requirements

The system uses ROS 2 Humble, PX4 SITL, and Gazebo Harmonic in Linux. The UAV supplies independent camera observations; the UGV consumes a fused map pose and guarded routes. Truth is confined to sensor synthesis and scoring. Positions use ENU map coordinates and body data use FLU. The verification library consumes normalized numerical traces independently of ROS and Gazebo.



R1 checks sampled planar disk clearance against static boxes/cylinders. R2 checks arrival within the declared deadline, with estimated and physical tolerances and a quiet dwell. R3 checks bounded response to an actual invalid guard, followed by a sustained zero command until recovery. R4 separates actuator-output expiry from independently measured physical settling. R6 checks acquisition-matched position error conditional on a valid pose and fresh UAV input. Availability follows the exported profiles and capability table. Planned exclusions are {"R5": "No hazard-event producer is available.", "R7_R8": "No online detector exists; enabled/disabled/forced-miss comparisons are unavailable.", "combinations": "Outage trials use goal A for empty/sparse and goal B for dense/corridor; both goals have nominal trials in every layout.", "seeds": "Seeds configure synthetic sensor corruption; PX4/render scheduling is not deterministic."}. No supervisor model checker is implemented.

Recorded-trace conditions. Let A denote a confirmed arrival dwell inside both tolerances with quiet commands; I, Q, V denote invalid guard, quiet command, and valid guard; and E denote fresh UAV input with valid pose.

$$\begin{aligned} R1: &\quad G_{\text{valid}}(d_{\text{clear}} \geq -\delta_C) \\ R2: &\quad F_{[0,D_A]}A \\ R3: &\quad G(I \rightarrow F_{[0,D_G]}(Q \wedge (Q\ W\ V))) \\ R6: &\quad G(E \rightarrow e_{\text{matched}} \leq \epsilon_L) \end{aligned}$$

Weak until W permits a quiet hold to continue to recording end without requiring recovery. R4 bounds actuator-output expiry and independently measured physical settling with a confirming dwell. The profile supplies the arrival deadline D_A, guard deadline D_G, error bound ϵ_L, dwell times, and all other numerical bounds; `bounds.csv` retains every evaluated parameter set. Source profiles and saved parameters remain authoritative.

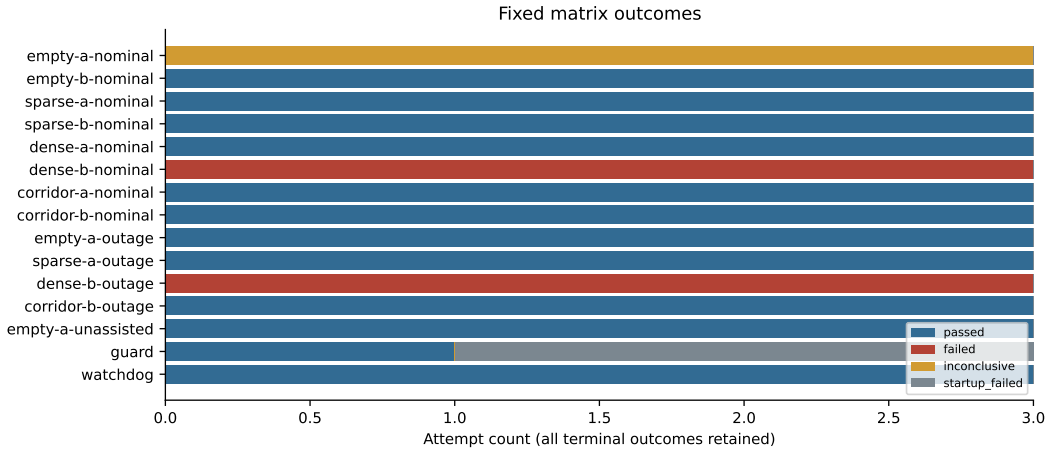
2 Temporal semantics and instrumentation

Integer nanosecond stamps retain clock domain and epoch. Acquisition, consumer receipt, command publication, actuator receipt/output, and truth motion remain separate reference points. Closed response windows include the deadline. Discrete samples expire at their maximum validity gap; interpolation is confined to continuous pose scoring within declared matching gaps. Missing coverage, untriggered obligations, and insufficient horizons remain inconclusive. A witnessed violation is retained despite unrelated incomplete evidence. R4 uses steady time and interval-average physical motion with sampling uncertainty. A paused simulator cannot establish physical settling from repeated unchanged positions.

3 Experiment design

The frozen matrix has 45 planned attempts and 45 attempted cases. The complete ledger retains 51 attempts and 52 saved result recordings across 4 source/profile/evaluator cohorts.

Recorded matrix factors are {"goals": [[3.0, 0.5], [4.0, 0.0]], "layouts": ["simple_goal_demo", "simple_goal_demo_corridor", "simple_goal_demo_dense", "simple_goal_demo_sparse"], "seeds": [631, 632, 633]}. Seeds configure injected sensor noise, not deterministic PX4 or renderer scheduling. The exported plan records paired comparisons, calibration separation, and exclusions. Predetermined seeds provide coverage, not a high-confidence probability guarantee. The assistance comparison table retains 3 seed/world/goal/outage pairs, both attempt outcomes, and available arrival values. Missing startup evidence prevents an assistance-benefit claim for that pair. All attempted outcomes, including startup failure, timeout, interruption, invalid data, and incomplete evidence, remain in the ledger. Deterministic fixture repetitions are repeatability checks.



Profiles: m2-camera/v1, m2-guard/v1, m2-watchdog/v1, m3-camera-unassisted/v1. Cohort set: 36ac8227d77abd42 (4 cohorts, 46 recordings); full IDs and exclusions: figure_sources.json and saved results.

4 Results

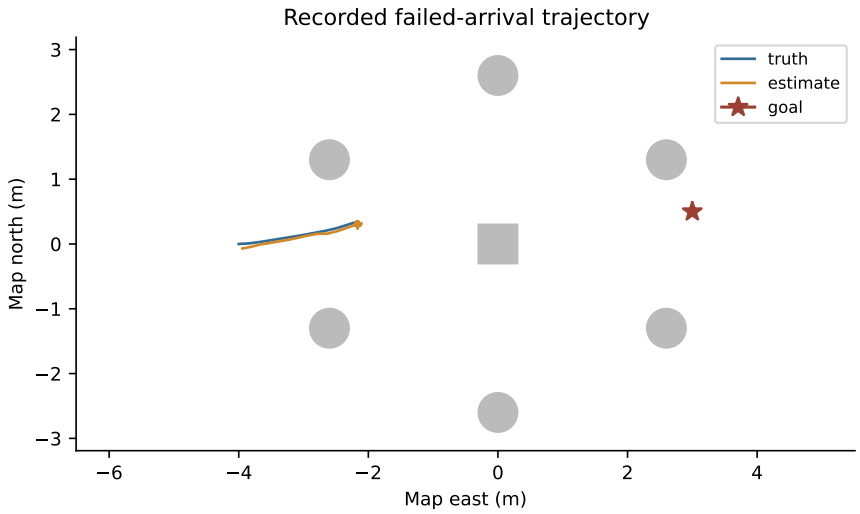
Unmeasured requirements: R5, R7, R8. Exact evaluated parameters are in `bounds.csv`.

Requirement	PASS	FAIL	Incomplete	Excluded	Triggers
R1	43	0	2	7	0
R2	34	8	3	7	43
R3	1	0	45	6	6
R4	6	0	0	46	6
R5	0	0	0	52	0
R6	39	0	3	10	40901
R7	0	0	0	52	0
R8	0	0	0	52	0

Counts refer to saved recordings; a watchdog attempt produces two separate recordings. Attempt-level denominators and missing recordings are retained separately in the outcome ledger. Totals are descriptive across matrix, search, and replay recordings in the listed cohorts, not a pooled success probability. Category counts are in `claims_by_category.csv`.

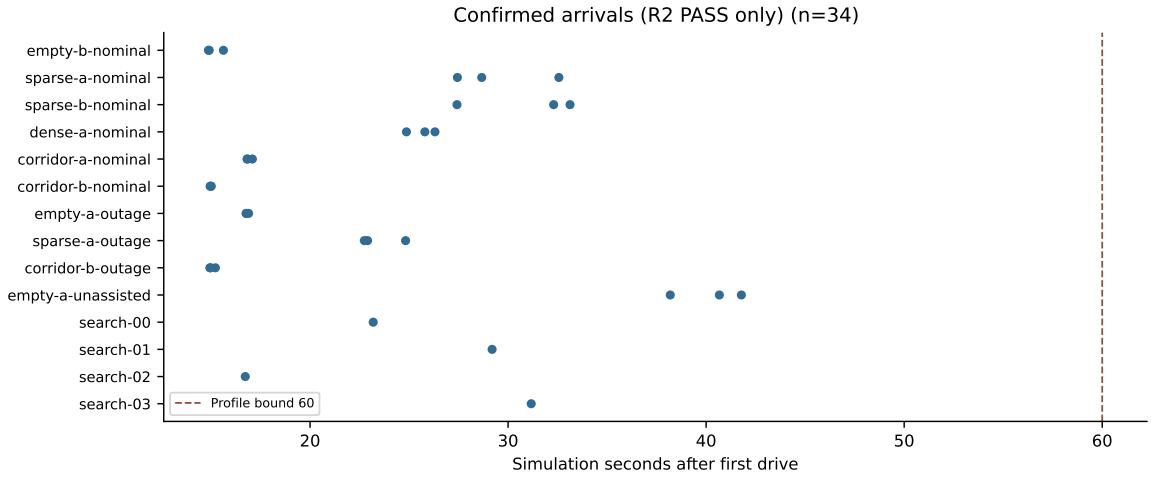
Measured metric	Recordings	Minimum	Maximum	Unit
minimum_clearance_m	33	0.121	0.480	m
arrival_s	34	14.868	41.780	simulation s
conditional_rmse_m	39	0.024	0.086	m
maximum_position_error_m	39	0.067	0.159	m
guard_response_ms	1	0.138	0.138	steady ms
watchdog_expiry_ms	6	366.106	381.620	steady ms
physical_settling_ms	6	426.636	454.561	steady ms

Metric counts include recordings with a measured value and a decided requirement verdict. All other attempted outcomes remain in the ledger. The complete witness list is in `violations.csv`; original failures remain alongside their fresh-world replays.



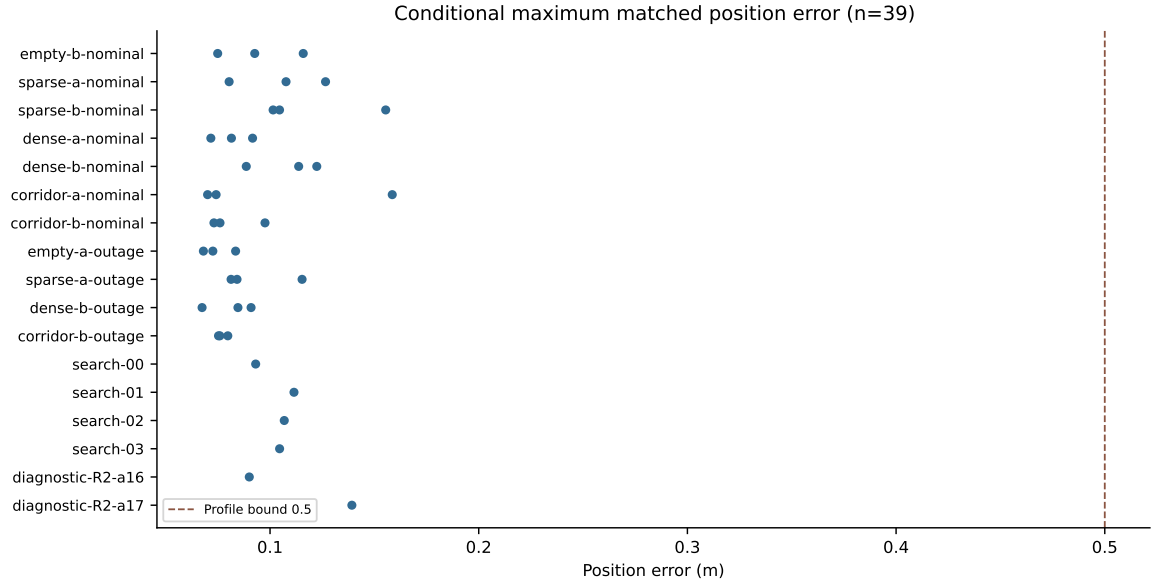
Profiles: m2-camera/v1. Cohort set: eb225c8fab610e83 (1 cohorts, 1 recordings); full IDs and exclusions: figure_sources.json and saved results.

The trajectory is a representative recorded trace with independently extracted collision geometry. Figure inputs are downsampled for display; all verdicts use full-resolution saved evaluations.



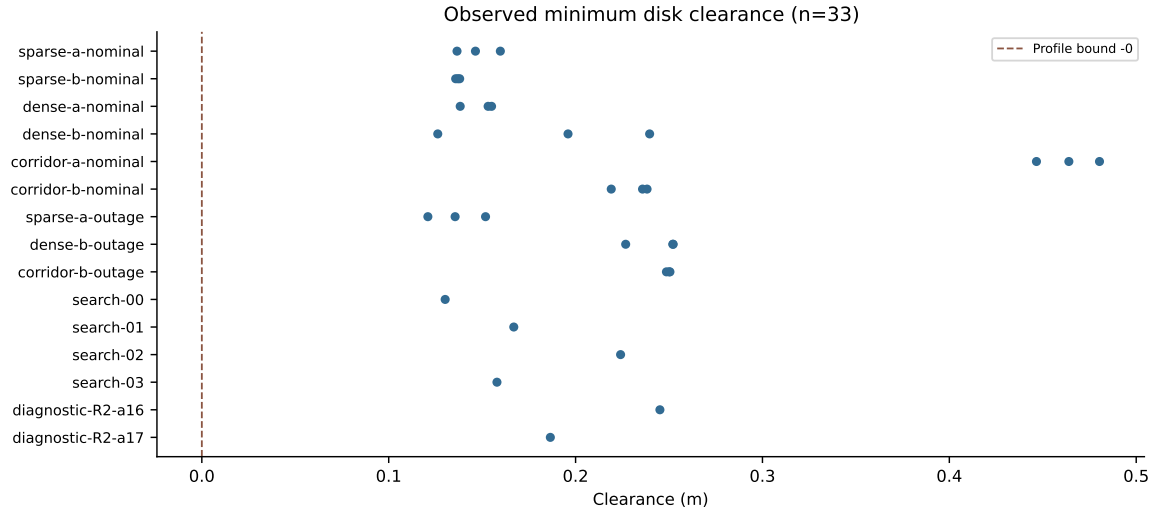
Profiles: m2-camera/v1, m3-camera-unassisted/v1. Cohort set: c7113a606e7034f4 (2 cohorts, 34 recordings); full IDs and exclusions: figure_sources.json and saved results.

Arrival values include R2 PASS recordings only. All other outcomes remain in the table and ledger. Physical and estimated arrival tolerances can differ; the profile-specific physical tolerance remains authoritative.



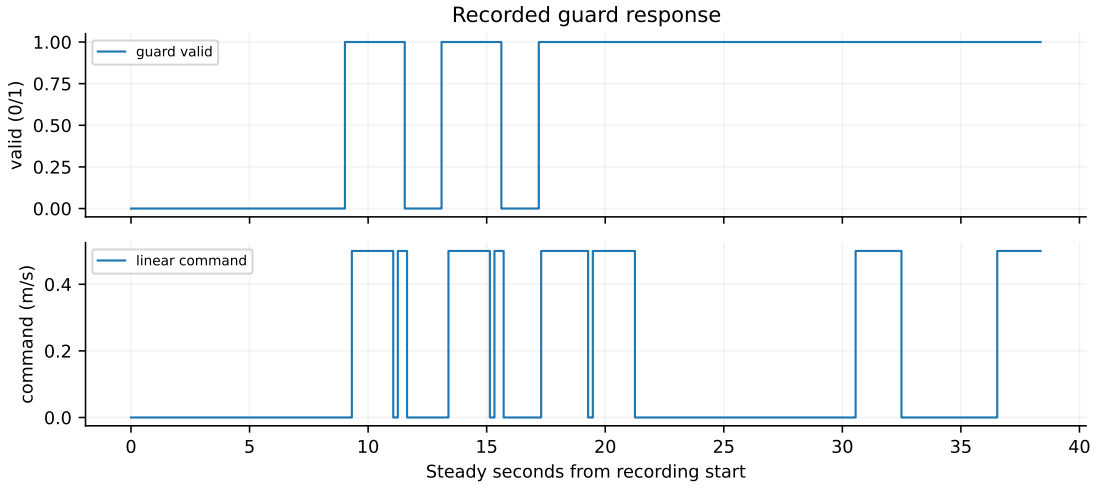
Profiles: m2-camera/v1. Cohort set: eb225c8fab610e83 (1 cohorts, 39 recordings); full IDs and exclusions: figure_sources.json and saved results.

Conditional position-error metrics retain guard/UAV eligibility, acquisition matching, and sample counts. They do not characterize unavailable UAV intervals or establish accuracy outside the recorded envelope.

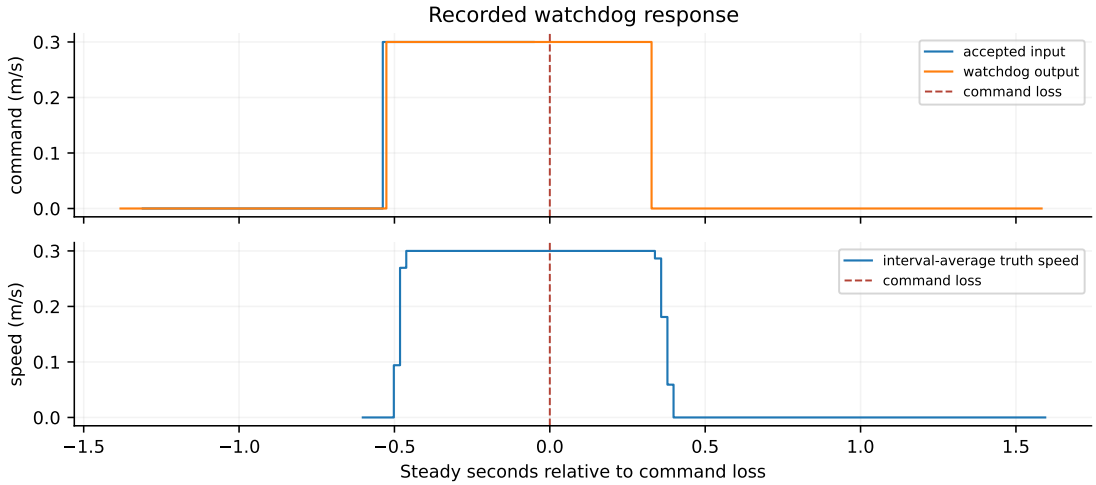


Profiles: m2-camera/v1. Cohort set: eb225c8fab610e83 (1 cohorts, 33 recordings); full IDs and exclusions: figure_sources.json and saved results.

Empty worlds have no finite nearest-obstacle clearance and do not supply a scalar search score. Negative clearance is a sampled disk-proxy overlap, distinct from exact rigid-body contact.



Profiles: m2-guard/v1. Cohort set: 2bcf52c6aa8d7898 (1 cohorts, 1 recordings); full IDs and exclusions: figure_sources.json and saved results.



Profiles: m2-watchdog/v1. Cohort set: 27460f6288a7d030 (1 cohorts, 1 recordings); full IDs and exclusions: figure_sources.json and saved results.

Consumer command response and watchdog output are distinct from measured physical settling. Authoritative timing metrics, including sampling uncertainty decisions, are retained in the saved results.

5 Failure search and replay

Search records 4 attempts and 274.824 measured driver wall seconds, with 0 search-counterexample replays and 2 supplemental diagnostic replays.

The sole search objective is minimum sampled R1 clearance. Missing evidence has no objective value. Generated worlds, offsets, seeds, commands, budgets, original failures, and replay links are retained. Matching seeds need not reproduce identical feedback trajectories. A finite search without a counterexample does not prove safety. Reproduced and non-reproduced outcomes remain separate. Post-hoc diagnostic replays remain separate from the held-out matrix and original search budget.

6 Claims, limits, and reproduction

The claim/evidence table records every requirement's verdict counts and trigger/eligible coverage. The results do not establish continuous collision freedom, online detector benefit, hazard response, or a discrete supervisor proof. Incompatible source/profile/evaluator cohorts remain separate. Calibration assumptions and evaluated

source-run identities remain explicit. Raw recording availability is explicit. Saved results and portable display series regenerate these tables and figures without simulation. The live demo reuses launch, recording, evaluation, and cleanup; mandatory failure or incomplete evidence prevents a successful verification summary. Recorded replay is labeled as replay.

Artifacts. See `metrics.csv`, `attempts.csv`, `claims.csv`, `figure_sources.json`, and the portable campaign ledger for exact numerical values, source hashes, profiles, commands, exclusions, and recording references.

Method references. Repository documents `docs/verification/requirements.md`, `interfaces.md`, and `experiments.md` define the executable requirements, finite-trace semantics, data contracts, and experiment protocol. The M2 evidence index records calibration conditions and bounds. This report covers the verification feature for integration into the project's ten-page report; it does not claim unimplemented project components.